



PROJECT PROPOSAL

Futsal Booking and Management System

B.Sc. CSIT | Tribhuvan University

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Introduction & Background

Current Landscape

Futsal Growth in Nepal

Rapid increase in urban futsal centers due to rising interest in indoor football and recreation. [1]

Manual Dependency

Most facilities rely on **phone calls**, **social media**, or **paper registers** for bookings.

Proposed Solution

Multi-Tenant SaaS Platform

Centralized web-based system allowing multiple businesses to operate independently on shared infrastructure.

Smart Innovation

Integration of **ML** for automated Sentiment Analysis

Problem Statement

Many futsal centers still use manual systems, which cause slow operations, booking mistakes, poor customer experience, and no proper business data insights.



Operational Chaos

Manual bookings via phone/social media cause **double bookings**, scheduling conflicts, and inaccurate record-keeping.



Discovery Barriers

No centralized platform for players to **discover and compare** multiple courts, prices, and reviews.



Business Intelligence Gap

Futsal owners lack proper data and smart tools, making it hard to track profits, popular times, customer satisfaction, or use AI insights

Project Objectives



Centralized Platform

To develop a unified web portal for **real-time discovery** and court booking, ensuring accessibility for all users.



Multi-Tenant SaaS

To implement a **scalable architecture** allowing multiple futsal facilities to operate independently within one system.



Community Features

To enable **verified ratings**, reviews, and discussion forums to enhance user interaction and trust.



AI/ML Integration

To integrate **sentiment analysis** for smarter business decision-making.

Literature Review

Suhaimi & Sahid (2021)

Research highlights the necessity of replacing manual logbooks with database-driven platforms to improve booking visibility and scheduling clarity.[1]

Fauzi et al. (2022)

Web-based systems significantly increase accessibility and customer reach while reducing the risks associated with handwritten rental processes. [4]

Modern System Design

Multi-tenant SaaS architecture is recommended for shared platforms to ensure proper data isolation and scalability for independent facility owners. [6]

Machine Learning Trends

Increasing application of Logistic Regression and TF-IDF for sentiment analysis and text feature extraction in user engagement platforms. [2][7]

Development Methodology

The project uses the Agile Development Model, which focuses on making progress step by step, getting regular feedback, and easily adjusting to changing needs.

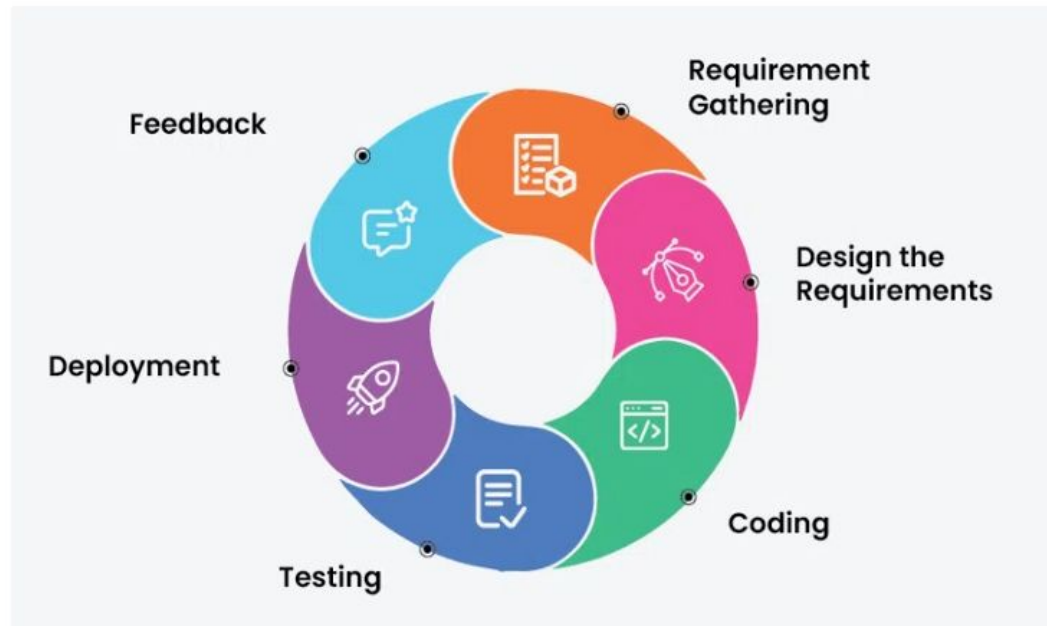


Fig: Agile Development Model [3]

Functional Requirements



End User

- ▣ Search futsal centers by **location**, **price**, and **slots**
- ▣ Real-time court booking and instant confirmation.
- ▣ Submit verified **ratings and reviews**.
- ▣ Personalized recommendations based on proximity.
- ▣ Manage booking history and profile settings.



Futsal Owner

- ▣ Manage futsal profile and Court availability.
- ▣ Set dynamic pricing and time slot configurations.
- ▣ Approve/Manage booking requests and schedules.
- ▣ Respond to customer queries and reviews.

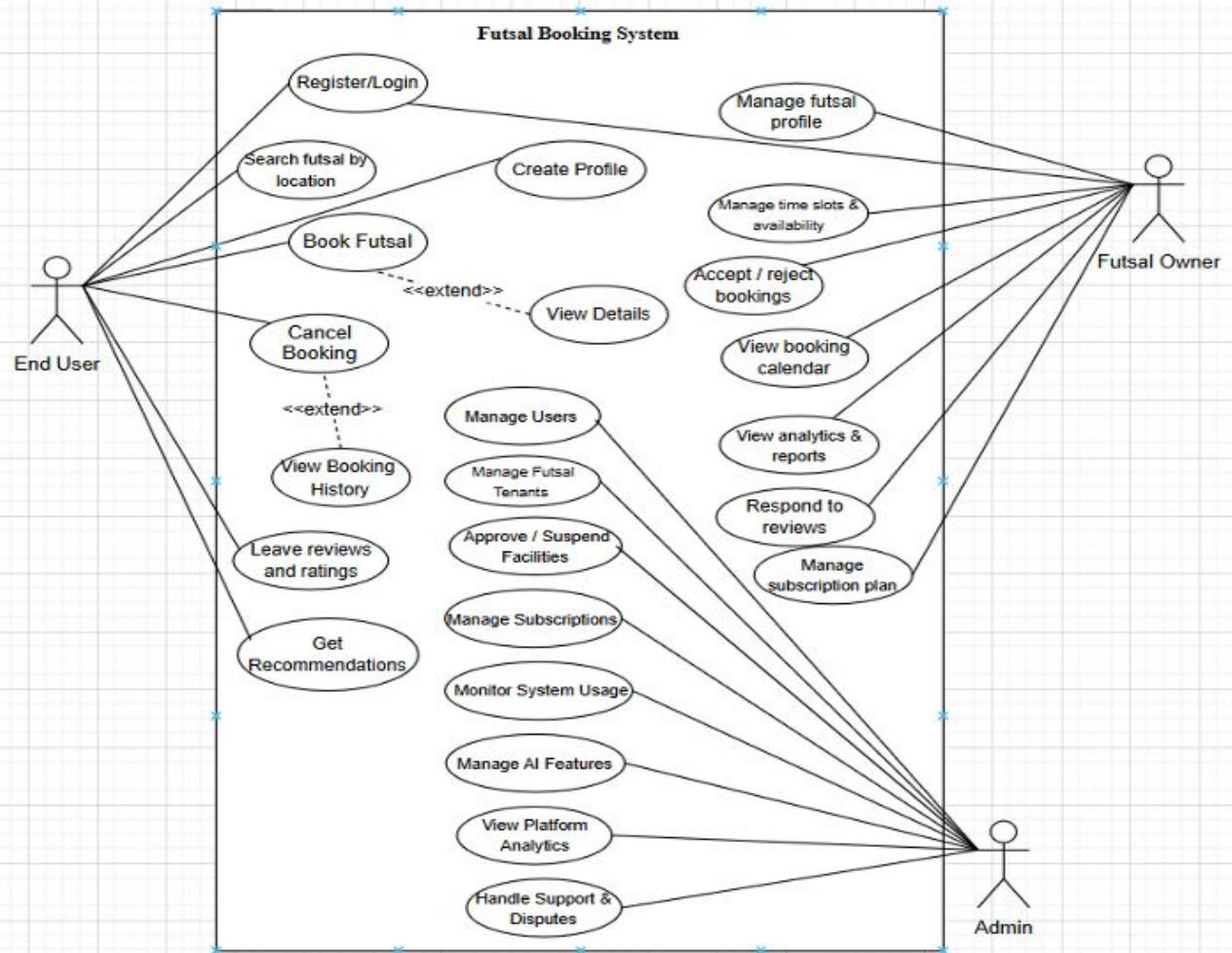


System Admin

- ▣ Manage all **multi-tenant accounts**.
- ▣ Monitor system-wide usage and generate reports.
- ▣ Monitor system-wide usage and package tiers..
- ▣ Moderate user accounts and content.
- ▣ System-level configuration and maintenance.

Use Case

Futsal Booking & Management System



Non-Functional Requirements



Performance

Ensuring **real-time availability** checks and high-speed response times for booking confirmations and slot validations.



Scalability

A scalable **multi-tenant architecture** designed to handle a growing number of futsal centers and concurrent users seamlessly.



Security

Implementation of **JWT authentication** and strict data isolation between tenants to ensure privacy and data integrity.



Usability

A **mobile-responsive** design ensuring a consistent and intuitive user experience across all devices and screen sizes.

Schedule Feasibility

Phase	Task	Duration (in weeks) 2082													
		1st	2nd	3rd	4th	5th	6th	7th	8th	9th	10th	11th	12th	13th	14th
1	Requirement Analysis														
2	System Design														
3	Docker Environment Setup														
4	Database and Redis Setup														
5	Backend Development														
6	Frontend Development														
7	ML Models Development														
8	System Integration														
9	Testing														
10	Documentation														

Technical Feasibility



Frontend

- **React.js** for dynamic UI
- Responsive design for mobile
- JWT-based session handling



Backend

- **Node.js & Express**
- Scalable RESTful API design
- Multi-tenant data isolation



Data Layer

- **MySQL** for persistence
- **Redis** for slot caching
- Cloud-based file storage



AI & ML

- **Python** NLP libraries
- Logistic Regression model
- TF-IDF text processing

Algorithms Description

Slot Conflict Prevention

The algorithm checks existing bookings for the selected court & date.

For a new booking request with start time S and end time E , the system verifies overlap with existing slots (S_i, E_i).

→ If overlap exists → Booking is rejected.

→ If no overlap → Booking is confirmed and slot is locked.

Mathematical Logic:

Two slots overlap if: $(S < E_i) \text{ AND } (E > S_i)$.

Booking is allowed only if:

$\forall i: \text{NOT}[(S < E_i) \text{ AND } (E > S_i)]$

This guarantees that only one active booking exists per court per time range

Location Distance (Haversine Formula)

The algorithm reads user latitude and longitude.

It compares them with futsal center coordinates.

Distances are calculated and sorted ascending.

Nearest futsal centers are recommended first.

Mathematical Logic:

$d = 2r \times \arcsin(\sqrt{(\sin^2(\Delta\phi/2) + \cos(\phi_1) \times \cos(\phi_2) \times \sin^2(\Delta\lambda/2))})$ Where:

d = distance between two points

r = Earth radius (6371 km)

ϕ = latitude, λ = longitude (in radians)

$\Delta\phi = \phi_2 - \phi_1$

$\Delta\lambda = \lambda_2 - \lambda_1$

Logistic Regression

TF-IDF feature vectors are generated from reviews. These features are passed into the logistic regression model. The model computes probability scores. A threshold (usually 0.5) is used to assign class labels.

→ Probability $\geq$ threshold → **Positive** sentiment

→ Probability $<$ threshold → **Negative** sentiment

Mathematical Logic:

$p = 1 / (1 + e^{(-z)})$

Where:

$z = w_0 + w_1x_1 + w_2x_2 + \dots + w_nx_n$

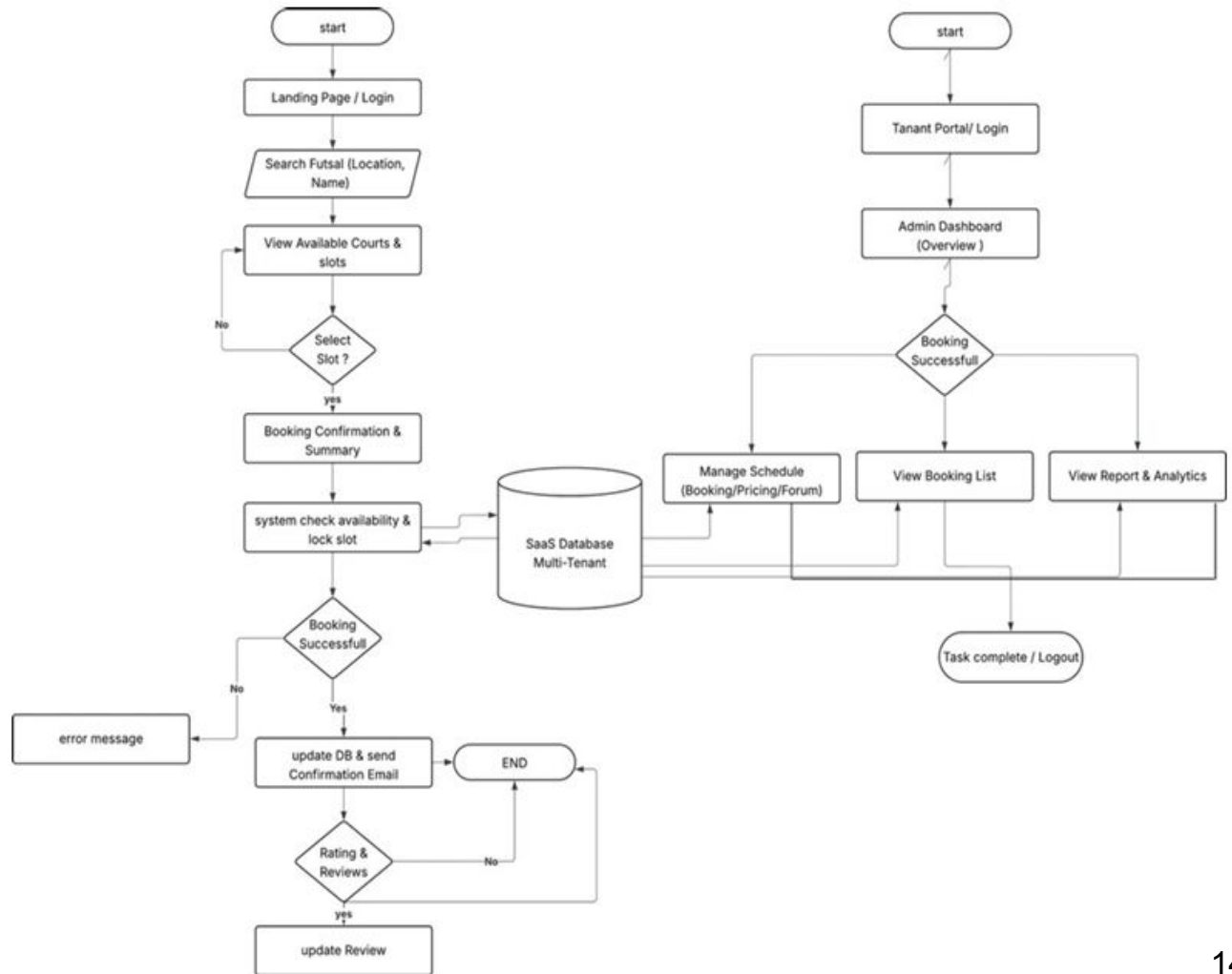
Decision rule:

Class = 1 if $p \geq 0.5$

Class = 0 if $p < 0.5$

Flow Chart

Futsal Booking & Management System



Expected Outcomes



Operational Efficiency

- Elimination of **double bookings** and manual scheduling errors.
- Automated revenue tracking and **business intelligence** reports.
- Streamlined customer communication and subscription management.



Enhanced Experience

- One-stop platform for **real-time discovery** and court comparison
- Convenient booking process with instant mobile confirmations.
- Access to **verified reviews** and proximity-based recommendations.



Scalable Innovation

- Robust **multi-tenant SaaS** model ready for commercial expansion.
- AI-driven insights through **sentiment analysis** of user feedback.
- Secure, high-performance infrastructure for nationwide futsal management.

References

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[5] SportsKey, "Sports Facility Booking Software Trends," 2026. Available: <https://www.sportskey.com>

[6] UiTM, "Progressive Web Applications for Mobile-First Systems," 2024.

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Thank You!

Futsal Booking and Management System

QUESTIONS?

Project Team



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